

1 Security Architecture Report

Architecture revision: REV-000003 | **Generation:** DOC-GEN-000003 | **Status:** Machine-generated review draft

Authority boundary: Architecture-level security advisory for human review. It does not certify controls, confirm control effectiveness, or grant procurement, installation, or customer-issue authority.

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2.1 Security Architecture

Section status:  Accepted | **Completeness:** Unresolved — engineering input required

The baseline security architecture is projected from the selected architecture zones and governed crossings. Architecture-level security zones, governed flows, constraints and findings were compiled for this revision. The result defines architecture-level boundaries and review actions; it is not compliance certification, control-effectiveness evidence, risk acceptance, or deployable configuration.

Advisory scope

- Applied technology overlays: 0
- Architecture/security findings: 26
- System guardrails: 4
- Selected baseline zones: 8
- Governed crossings: 7
- Security architecture guidance patterns: 16 (reference catalogue; not selected topology or deployed enforcement)
- Planning capabilities: 15 (unquantified; not procurement-ready)
- Security Architect handoffs: 25

Architecture/security finding severity

High	Medium	Low
20	5	0

System guardrails

The following 4 critical system guardrails are enforced. They are product boundaries, not critical vulnerabilities in the proposed customer architecture.

- Guardrail 1: Compliance, certification and control effectiveness require separate evidence and review.
- Guardrail 2: Residual risk acceptance remains a human decision.
- Guardrail 3: Architecture guidance does not constitute deployable device, firewall, switch or credential configuration.
- Guardrail 4: Governed architecture facts remain read-only during security assessment.

Baseline security zones — selected architecture

Zone	Architecture purpose	Source authority
zone.compute	Compute Fabric Zone	Selected architecture baseline
zone.external	External / Customer Zone	Selected architecture baseline
zone.management	Management Zone	Selected architecture baseline
zone.oob	OOB Management Zone	Selected architecture baseline
zone.operations	Operations / Logging Zone	Selected architecture baseline
zone.platform	Platform Services Zone	Selected architecture baseline
zone.storage	Storage Zone	Selected architecture baseline
zone.tenant_workload	Tenant Workload Zone	Selected architecture baseline

Governed baseline crossings

Source	Destination	Flow class	Disposition
zone.management	zone.oob	privileged_administration	unresolved_detailed_design
zone.compute	zone.storage	storage_data_path	architecturally_required
zone.external	zone.platform	identity_assertion	unresolved_external_dependency
zone.platform	zone.compute	control_plane	architecturally_required
zone.storage	zone.operations	telemetry	architecturally_required
zone.compute	zone.operations	telemetry	architecturally_required
zone.external	zone.platform	workload_submission	architecturally_required_controls_unresolved

Explicit-allow and deny-unless-required values are security architecture guidance. They do not prove current deployed permission, prohibition, or control effectiveness.

Architecture findings

Finding	Severity	Domain	Required architecture action
Finding 1	High	oob	OOB management must be represented as a distinct restricted security zone separated from user/workload access.
Finding 2	High	oob	Switch management interfaces must be isolated from workload and user access paths.
Finding 3	High	oob	User-plane reachability to IPMI/BMC management services must be denied by architecture.
Finding 4	High	oob	Any OOB route handoff to customer networks must be explicit, bounded, and identified as privileged management exposure.
Finding 5	Medium	oob	The architecture assessment must report whether OOB management has redundant uplinks/paths when upstream facts provide them.
Finding 6	Medium	oob	OOB administrative access and device-management events require a security logging/monitoring handoff.

Finding	Severity	Domain	Required architecture action
Finding 7	High	control_plane	Observability backends containing infrastructure telemetry must be placed in a restricted monitoring zone.
Finding 8	High	control_plane	Administrative identity, MFA/PAM and break-glass requirements must be emitted as implementation handoffs; The architecture assessment does not assert effectiveness.
Finding 9	Medium	control_plane	Unknown platform/operator version support must remain unresolved rather than assumed compatible.
Finding 10	High	fabric	Compute-fabric access must be limited to workload communication and authorized management paths; external/user ingress must not be inferred.
Finding 11	High	fabric	Where upstream design distinguishes storage and compute fabrics, The architecture assessment must preserve those boundaries rather than flattening them.
Finding 12	High	fabric	NVLink/NVSwitch scale-up domain must be represented as a distinct high-trust GPU

Finding	Severity	Domain	Required architecture action
Finding 13	High	tenancy	communication domain and not exposed as routable user network. Unknown tenancy mode must prevent automatic selection of single-tenant or multi-tenant security overlays.
Finding 14	High	confidential	GPU confidential-computing capability must remain unknown unless the hardware SKU/capability is evidenced upstream.
Finding 15	High	storage	High-performance storage must be represented as a distinct data/security zone with explicit compute-to-storage flows.
Finding 16	High	storage	Storage administrative interfaces must terminate in privileged management zones and not user/workload zones.
Finding 17	Medium	storage	Unknown encryption-at-rest state must remain unresolved and cannot be reported satisfied.
Finding 18	High	storage	Storage authentication credentials/keys are implementation artifacts and must

Finding	Severity	Domain	Required architecture action
			never be generated by the security architecture assessment.
Finding 19	High	external	Customer edge/uplink connectivity must be represented as an external trust boundary with explicit route/flow classes.
Finding 20	High	external	Unknown external exposure must remain unresolved and visible in the architecture assessment output.
Finding 21	High	observability	Security and infrastructure telemetry stores must be represented as restricted services.
Finding 22	High	observability	Administrative actions require an audit/telemetry handoff; The architecture assessment reports architecture requirement not logging effectiveness.
Finding 23	Critical	identity	Administrative identity roles must be distinct from ordinary user/workload identities.
Finding 24	High	identity	Emergency/break-glass administrative access must be a governed implementation handoff, not auto-

Finding	Severity	Domain	Required architecture action generated by the security architecture assessment.
Finding 25	High	identity	Unknown MFA/PAM state must remain unresolved; platform support does not prove deployment.
Finding 26	Medium	supply_chain	Version-specific NVIDIA reference-architecture support claims require explicit version facts; otherwise remain unresolved.

Required Security Architecture Capabilities — Planning Only

These entries identify capabilities required or recommended by the compiled security architecture. They do not assert that the capabilities are deployed, licensed, configured, effective or quantitatively sized.

- OOB management segmentation — quantity unresolved; procurement and installation authority: No.
- Admin control-plane isolation — quantity unresolved; procurement and installation authority: No.
- User control-plane boundary — quantity unresolved; procurement and installation authority: No.
- BlueField root-of-trust/secure management — quantity unresolved; procurement and installation authority: No.
- BlueField attestation — quantity unresolved; procurement and installation authority: No.
- GPU attestation — quantity unresolved; procurement and installation authority: No.
- Attestation policy/key broker services — quantity unresolved; procurement and installation authority: No.
- Kata confidential sandbox — quantity unresolved; procurement and installation authority: No.
- Compute/management fabric segmentation — quantity unresolved; procurement and installation authority: No.
- Cross-plane tenant isolation — quantity unresolved; procurement and installation authority: No.
- Security observability/audit — quantity unresolved; procurement and installation authority: No.
- Artifact provenance policy — quantity unresolved; procurement and installation authority: No.

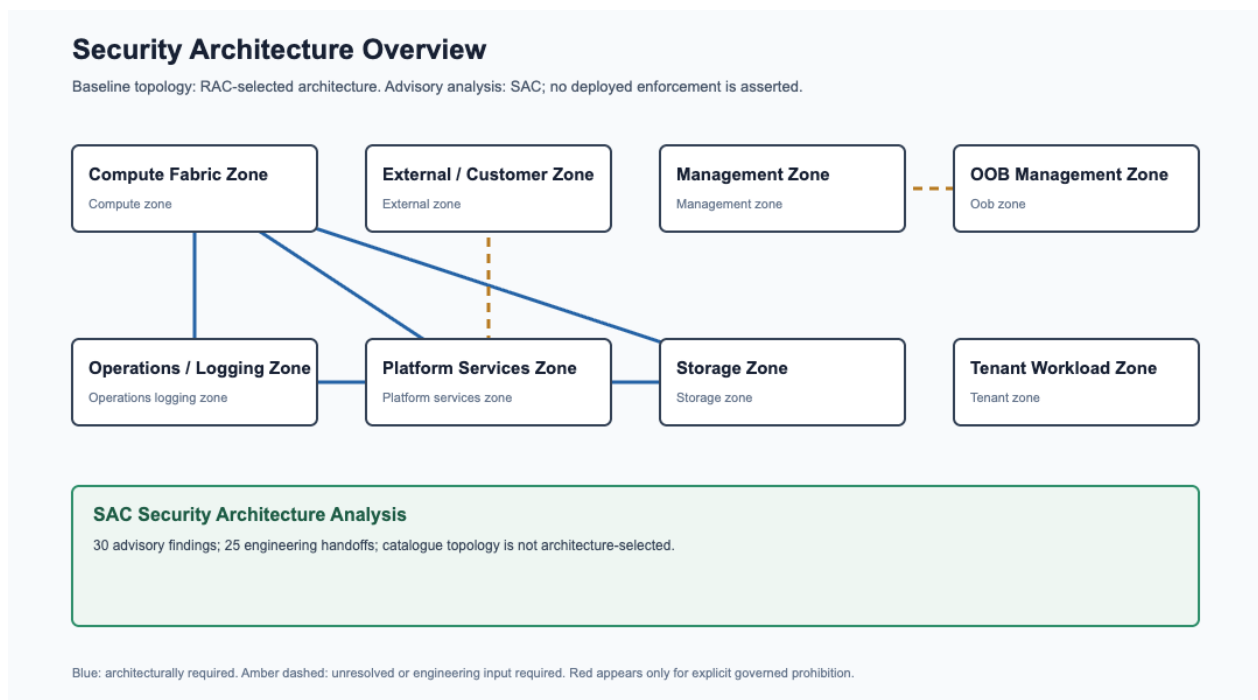
- External boundary controls — quantity unresolved; procurement and installation authority: No.
- Storage identity/isolation — quantity unresolved; procurement and installation authority: No.
- Privileged/user/service identity separation — quantity unresolved; procurement and installation authority: No.

Security engineering handoff

- Action 1: OOB management must be represented as a distinct restricted security zone separated from user/workload access.
- Action 2: Switch management interfaces must be isolated from workload and user access paths.
- Action 3: User-plane reachability to IPMI/BMC management services must be denied by architecture.
- Action 4: Any OOB route handoff to customer networks must be explicit, bounded, and identified as privileged management exposure.
- Action 5: OOB administrative access and device-management events require a security logging/monitoring handoff.
- Action 6: Observability backends containing infrastructure telemetry must be placed in a restricted monitoring zone.
- Action 7: Administrative identity, MFA/PAM and break-glass requirements must be emitted as implementation handoffs; The architecture assessment does not assert effectiveness.
- Action 8: Unknown platform/operator version support must remain unresolved rather than assumed compatible.
- Action 9: Compute-fabric access must be limited to workload communication and authorized management paths; external/user ingress must not be inferred.
- Action 10: Where upstream design distinguishes storage and compute fabrics, The architecture assessment must preserve those boundaries rather than flattening them.
- Action 11: NVLink/NVSwitch scale-up domain must be represented as a distinct high-trust GPU communication domain and not exposed as routable user network.
- Action 12: Unknown tenancy mode must prevent automatic selection of single-tenant or multi-tenant security overlays.
- Action 13: GPU confidential-computing capability must remain unknown unless the hardware SKU/capability is evidenced upstream.
- Action 14: High-performance storage must be represented as a distinct data/security zone with explicit compute-to-storage flows.
- Action 15: Storage administrative interfaces must terminate in privileged management zones and not user/workload zones.
- Action 16: Unknown encryption-at-rest state must remain unresolved and cannot be reported satisfied.
- Action 17: Storage authentication credentials/keys are implementation artifacts and must never be generated by the security architecture assessment.
- Action 18: Customer edge/uplink connectivity must be represented as an external trust boundary with explicit route/flow classes.

- Action 19: Unknown external exposure must remain unresolved and visible in the architecture assessment output.
- Action 20: Security and infrastructure telemetry stores must be represented as restricted services.
- Action 21: Administrative actions require an audit/telemetry handoff; The architecture assessment reports architecture requirement not logging effectiveness.
- Action 22: Administrative identity roles must be distinct from ordinary user/workload identities.
- Action 23: Emergency/break-glass administrative access must be a governed implementation handoff, not auto-generated by the security architecture assessment.
- Action 24: Unknown MFA/PAM state must remain unresolved; platform support does not prove deployment.
- Action 25: Version-specific NVIDIA reference-architecture support claims require explicit version facts; otherwise remain unresolved.

2.2 Figure 1: Security Architecture Overview



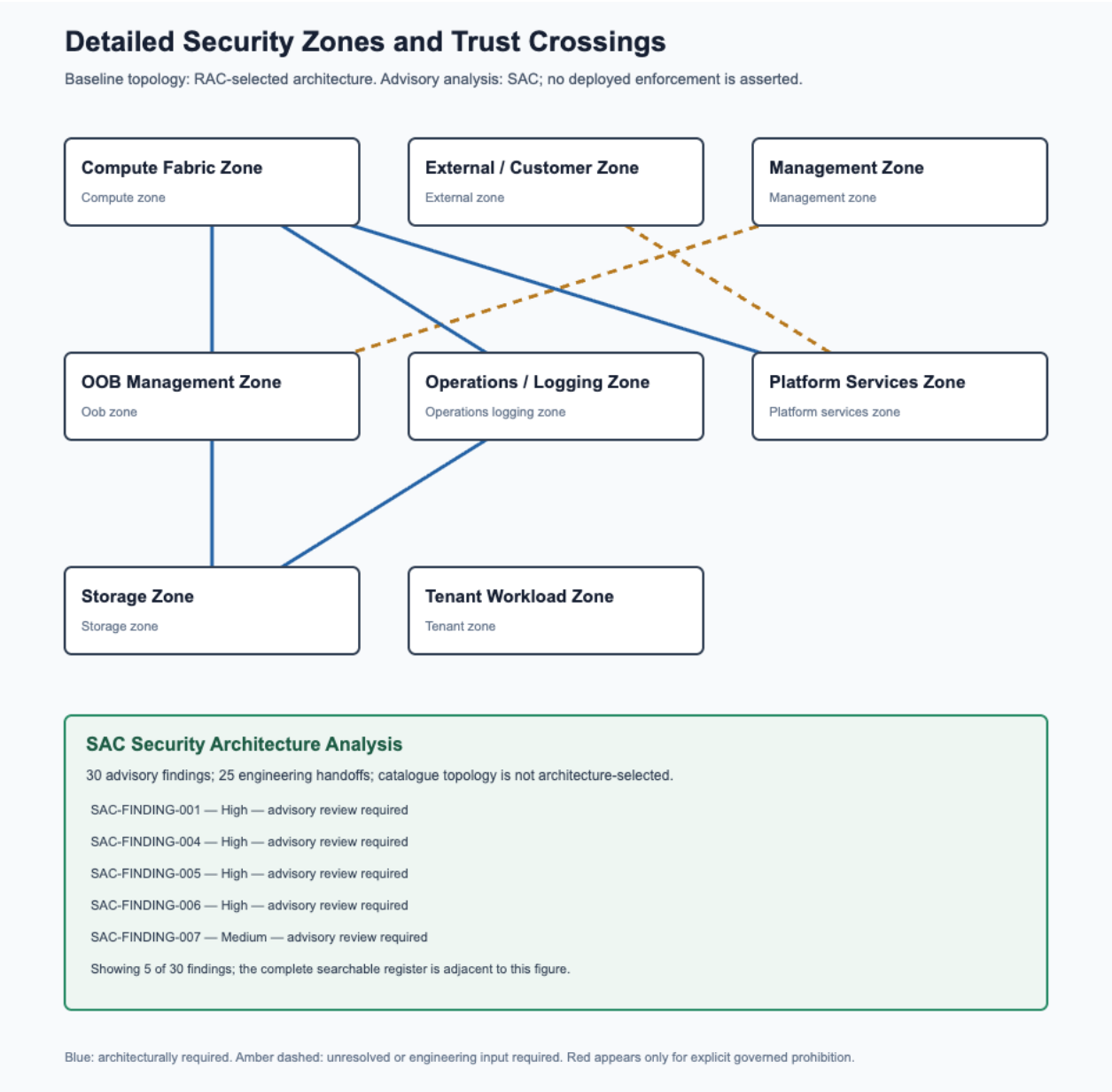
Security Architecture Overview

Purpose: Security Architecture Overview

How to read this figure: This Security Architecture Overview presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.3 Figure 2: Security Zones and Governed Crossings



Security Zones and Governed Crossings

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